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Contribution Journal

[Min Ma Kha]

My role in Lab 03 revolved around both organizational roles and analytical contributions. To ensure all members of the group were able to conduct the experiment independently, I setup sharing the group notebook, after which I managed the submission of our lab materials on the GitHub repository. Besides, I prepared both the reflection journal and the contribution journal template to ensure consistent format while submitting. My assigned section is the completion of cells 21 to 25, where I conducted the assessment of machine learning types, provided case studies, and the analysis of results.

For cell 24, I provided a critical analysis of results by indicating that Logistic Regression achieved 88.9% accuracy while Decision Tree only reached 83.3%, hence I conducted additional feature experiments that improved the model's performance to 91.7% trial. Using the confusion matrix, I was capable of identifying the cases where the models failed, and I noted that both models classified class_0 correctly but the Logistic Regression made an error in which three class_2 wines were incorrectly classified as class_1.

The fact that I made my selection of the best features based on test score indicates its limitation, thus, cross-validation is required to make sure that the result is valid.

[Chris Roy]

I had worked on running and understanding how to use the Sklearn library as well as the matplotlib library in tandem to present datasets in a visually pleasing manner. I had worked on cells 6 through 10 to understand how to load wine and unpack all its information and display it on a plotted graph.

[Camille Casas]

Cells 11–12: Model Evaluation and Interpretation

My contribution focused on examining how well the trained machine-learning model performed rather than simply relying on the final accuracy score. I worked with the model's predictions and the confusion matrix to distinguish between correct classifications and misclassifications. I interpreted the diagonal values as correct predictions and the off-diagonal values as errors. This helped me understand that model performance should be examined at the individual class level, because a model can have relatively strong overall accuracy while still struggling to distinguish a particular wine class.

Cell 13: Feature Selection Experiment

I contributed to the hands-on feature-selection portion of the Wine Classification Project by selecting three specific features from the available Wine dataset variables: alcohol, color intensity, and proline. I used these selected variables to create a new feature set and then divided the data into training and testing sets. This experiment allowed me to investigate how limiting a model to only certain characteristics can influence its ability to classify the wines correctly.

Cell 14: Training My Logistic Regression Model

Using my selected features, I built and trained a new Logistic Regression classification model. I kept the training and testing process consistent so that my results could be meaningfully compared with the original model. This portion of my contribution reinforced my understanding that the algorithm itself is only one component of machine learning; the variables supplied to the algorithm can substantially influence the quality of its predictions.

Cell 15: Accuracy Comparison and Analysis

I evaluated my newly trained model by generating predictions for the testing data and calculating its accuracy. My model, using alcohol, color intensity, and proline, achieved approximately 83.3% accuracy, compared with approximately 88.9% accuracy for the original model. Therefore, my feature selection did not improve the original model's performance and resulted in a decrease of approximately 5.6 percentage points. Rather than treating this as an unsuccessful contribution, I used the result to determine that the original feature configuration contained information that was more useful for classification than my selected three-feature combination.

Overall Contribution:

Through Cells 11–15, I contributed to both the evaluation and experimental stages of the machine-learning workflow. My work involved interpreting classification performance, examining a confusion matrix, selecting alternative features, training a Logistic Regression model, generating predictions, and comparing the resulting accuracy against the original model. The most valuable finding from my contribution was that reducing or changing features does not automatically improve a model. My results demonstrated why feature selection should be treated as an experimental process in which performance is measured and compared rather than assumed.

[Rigoberto Padilla]

I worked on Cells 1–5 of the Lab 03 notebook. I focused on understanding the main objectives of the lab and the three different types of machine learning: supervised, unsupervised, and reinforcement learning. I also ran the library setup in Cell 4 and made sure the notebook was ready for the rest of the lab. The first part helped me understand what we were going to be doing before getting into the actual dataset and models. I also went through the Wine dataset introduction so I had a better idea of what we would be working with later in the notebook.

[Samantha Mireles]

For my part of the group project, I worked on Cells 16–20. I reviewed the different types of data used in machine learning and worked on the hands-on feature experiment. For my experiment, I chose alcohol, hue, and proline as the three features from the Wine dataset. I ran the Logistic Regression model with those features and got an accuracy of about 97.2%. This helped me understand that the features we choose can have a big effect on how well a model performs. I also made sure my assigned cells ran correctly, checked the results, and shared what I found with my group.